

# Layer-2 Overlay Network over WireGuard

0.5cm] VPS Proxmox Virtualization Environment

1.5cm]

Technical Architecture & Implementation Document

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Year: 2026

3cm]

# Contents

## Revision History

| Date       | Version | Description                                               |
|------------|---------|-----------------------------------------------------------|
| 2026-02-15 | 1.0     | Initial release of full technical documentation.          |
| 2026-02-16 | 1.1     | Added diagrams, shaded headers, sidebars, skills section. |

## Skills Demonstrated

- Advanced Linux networking (WireGuard, GRE-TAP, bridges)
- Secure tunnel design and encryption best practices
- Proxmox virtualization and network architecture
- Systemd automation and service orchestration
- Debugging complex distributed systems
- Packet analysis using tcpdump and iproute2
- NAT, routing, and overlay network design
- Technical documentation and architecture writing

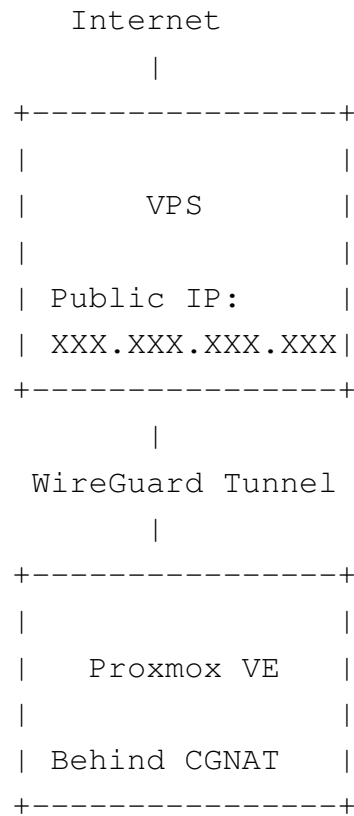
## Executive Summary

This document describes the design and implementation of a secure, encrypted Layer-2 (L2) overlay network between a public VPS and a Proxmox hypervisor hosted on a private FTTH connection. The solution uses WireGuard as the Layer-3 transport, GRE-TAP for Ethernet encapsulation, and Linux bridges to merge both sites into a single broadcast domain.

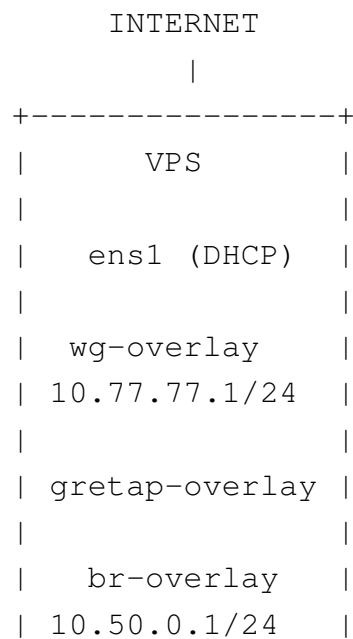
The overlay enables virtual machines on Proxmox to behave as if they were physically located inside the VPS network, supporting ARP, DHCP, multicast, and broadcast traffic. The system is automated using systemd and is production-ready.

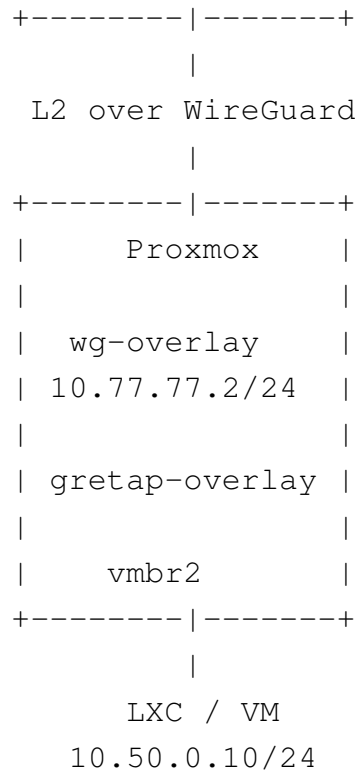
## Architecture Overview

### Logical Diagram

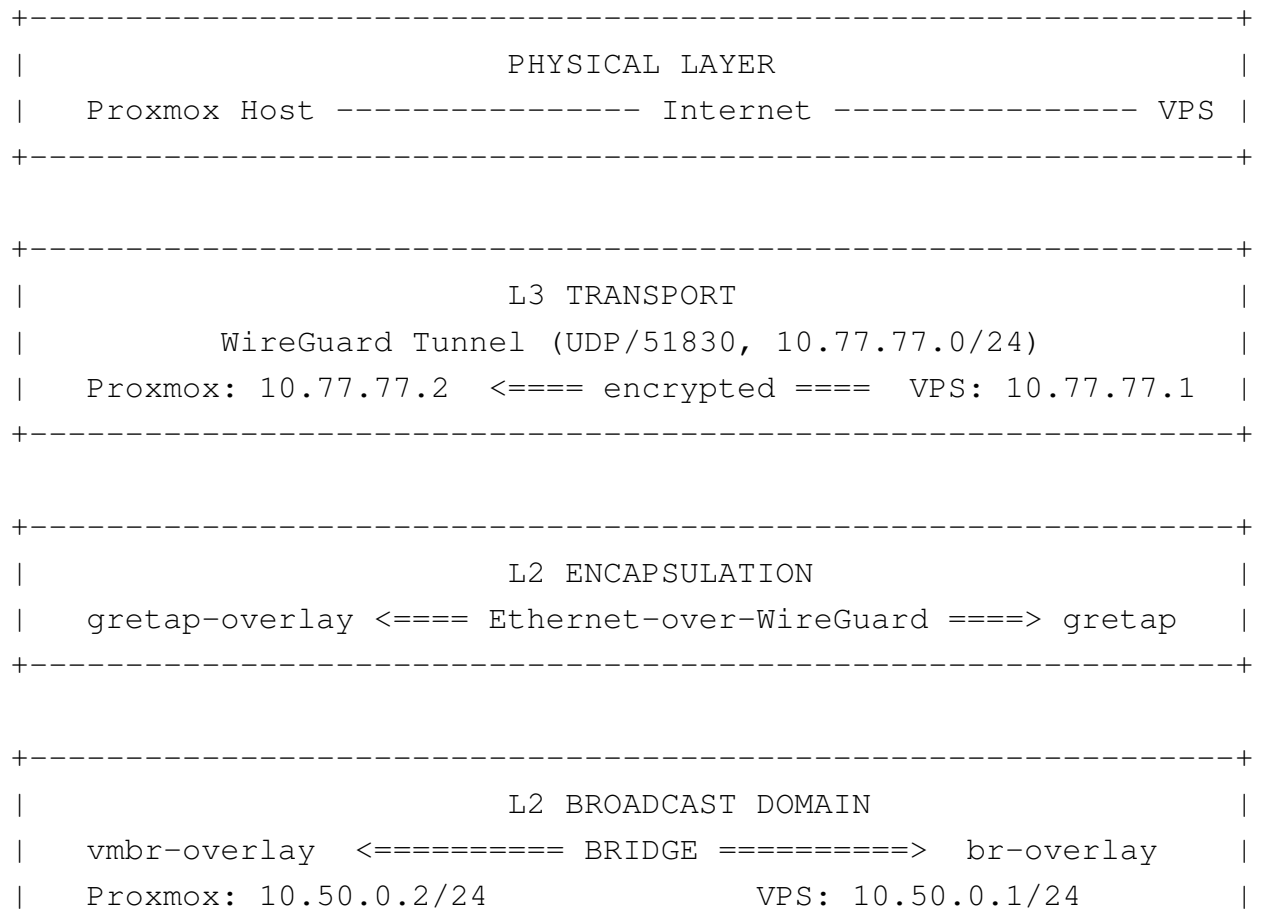


### Hybrid L2 Overlay over WireGuard





## Full Multi-Layer Diagram



|         |                                                           |  |
|---------|-----------------------------------------------------------|--|
| +-----+ |                                                           |  |
| +-----+ |                                                           |  |
|         | VIRTUAL MACHINES                                          |  |
|         | VM / LXC Containers (10.50.0.x) behave as if local to VPS |  |
| +-----+ |                                                           |  |

## Implementation

### VPS Configuration

#### WireGuard

```
[Interface]
Address = 10.77.77.1/24
ListenPort = 51830
PrivateKey = <private>

PostUp = sysctl -w net.ipv4.ip_forward=1
PostDown = sysctl -w net.ipv4.ip_forward=0

[Peer]
PublicKey = <Proxmox-public>
AllowedIPs = 10.77.77.2/32
```

#### GRETAP

```
ip link add gretap-overlay type gretap \
    remote 10.77.77.2 local 10.77.77.1 dev wg-overlay
ip link set gretap-overlay up
```

#### Bridge

```
ip link add br-overlay type bridge
ip link set br-overlay up
ip link set gretap-overlay master br-overlay
ip addr add 10.50.0.1/24 dev br-overlay
```

# Proxmox Configuration

## WireGuard

```
[Interface]
Address = 10.77.77.2/24
PrivateKey = <private>

[Peer]
PublicKey = <VPS-public>
Endpoint = <VPS-IP>:51830
AllowedIPs = 10.77.77.1/32
PersistentKeepalive = 25
```

## GRETAP

```
ip link add gretap-overlay type gretap \
    remote 10.77.77.1 local 10.77.77.2 dev wg-overlay
ip link set gretap-overlay up
```

## Bridge

```
ip link add vmbr-overlay type bridge
ip link set vmbr-overlay up
ip link set gretap-overlay master vmbr-overlay
ip addr add 10.50.0.2/24 dev vmbr-overlay
```